

Our team aims to develop a collaborative airborne surveillance system using a swarm of runway independent Unmanned Aerial System (UAS) called **VAJRA-NET** (VTOL Adaptive Joint Radar Architecture Network). Unlike conventional airborne radar systems like the AWACS that rely on a single vulnerable command aircraft, VAJRA-NET proposes a resilient and self-healing distributed multistatic radar system capable of maintaining operational continuity even in contested and electronically challenging environments.

The architecture combines the principles of electronically steerable AESA-inspired radar systems with a multistatic radar sensing. Instead of equipping every drone with a complete radar payload, the swarm employs a role-based operational hierarchy consisting of a Command Node, Relay Nodes and Sentry Nodes. The primary sensing layer consists of 4 Sentry UAVs, where one UAV acts as the active transmission (TX) node and the other 3 act as the passive receiver (RX) nodes. The TX node carries a lightweight 4-element electronically steerable antenna array integrated with digital phase shifters which are controlled through an FPGA-based processing layer. By dynamically altering the phase of the transmitted waves at each of the antenna, the radar beam can be electronically steered across a wide angular sector at high speed without requiring mechanical antenna movement. To overcome the FOV of a compact field array, the TX UAV can perform a slow controlled yaw rotation (if needed) to enable an even greater FOV. This enable a continuous wide surveillance coverage while minimizing payload complexity and weight.

The RX Sentry nodes do not emit radar energy, thereby reducing power consumption and electronic detectability. Instead, they receive reflected radar signals from aerial targets illuminated by the TX node. Observations collected from multiple spatially distributed receiver UAVs are combined collaboratively to improve target localization accuracy, low Radar Cross Section (RCS) detection capability, and multi-target tracking performance. To ensure survivability and redundancy, one additional RX Sentry also carries dormant TX hardware capability, allowing it to immediately assume transmission duties in the event of failure or destruction of the primary TX node.

The Command Node operates from a safer rearward position within the swarm and is responsible for generating the consolidated Air Situation Picture (ASP), target fusion, mission-level coordination, and decision-making. Between the Sentry layer and the Command Node are dedicated Relay Nodes that maintain encrypted high-bandwidth communication pathways and ensure stable data transmission across the swarm network. This layered architecture minimizes communication bottlenecks while enabling scalable operations over larger operational areas.

A major innovation of VAJRA-NET lies in its decentralized and dynamically reconfigurable operational philosophy. Although each UAV initially performs a designated role, every node within the swarm possesses an identical modular hardware stack comprising a flight controller, companion computer, communication module, and FPGA-enabled reconfigurable compute layer. This allows dynamic reassignment of operational roles during mission execution. In the event of loss of the Command Node, the nearest Relay Node autonomously assumes command and fusion responsibilities. Similarly, if a Relay Node is lost, another available node dynamically expands its communication coverage to absorb and support the disconnected Sentry layer. The swarm continuously reorganizes itself using distributed heartbeat monitoring and autonomous role reassignment protocols, ensuring graceful degradation, self-healing capability, and uninterrupted mission continuity without dependence on a single vulnerable “mother drone.”

VAJRA-NET aims to provide a scalable indigenous pathway towards future distributed airborne surveillance and Counter-UAS systems while aligning closely with the operational objectives and evaluation criteria of Mehar Baba Competition-3.